

Summer term 2026

Visual Data Analysis

Assignment Sheet 11

Solution has to be submitted via eCampus by July 3, 2026, 10:00 a.m.

If you have questions concerning the exercises, please use the forum on eCampus.

- Please work on this exercise in **small groups** of 3 students. Submit each solution only once, but clearly indicate who contributed to it by forming a team in eCampus. Remember that all team members have to be able to explain all answers.
- Please submit your answers in PDF format, and your scripts as *.py/*.ipynb files. If you are using [Jupyter notebook](#), please also export your scripts and results as PDF.

Exercise 1 (Numerical Solution of the Volume Rendering Integral, 5 Points)

The volume rendering integral can be approximated as the following Riemann sum:

$$\sum_{i=1}^n \tilde{C}_i \prod_{j=1}^{i-1} (1 - A_j) \quad (1)$$

- For which of the quantities in Eq. (1) do the values change when performing lighting (e.g., Phong shading)? (1P)
- The definition of opacity $A := 1 - e^{-\kappa \cdot \Delta s}$ results from discretizing the light attenuation term with absorption coefficient κ and step size Δs . When changing Δs to another step size $\Delta s'$, A needs to be updated to A' correspondingly, so that it still amounts to the same absorption coefficient κ . Please derive an equation for the new opacity A' as a function of A , Δs , and $\Delta s'$ (3P). Will increasing Δs increase or decrease the corresponding A ? (1P)

Exercise 2 (Accelerated Volume Rendering, 5 Points)

- Pre-integration in direct volume rendering generates a two-dimensional lookup table that has to be re-computed when the transfer function changes. However, when the transfer function only changes in an interval $[l, r]$, parts of the lookup table remain unaffected. Denote the axes of the lookup table as f and b , for front and back values of a given integration step. Please specify a condition on l, r, f, b that evaluates to **True** exactly if the given position in the lookup table is affected. (2P) Also provide a sketch that highlights the affected region of the lookup table. (1P)
- Chapter 10, Slide 84 stated the ratio of inner octree nodes to original data points for a full octree in the optimal case of $s \times s \times s$ data points with $s = 2^n$. Compute the optimal ratio for 2D quadrees in the same way. (2P)

Hint: The derivation on the slide uses a summation formula for powers of eight:

$$\sum_{i=0}^{k-1} 8^i = \frac{8^k - 1}{7} \quad (2)$$

A very similar formula exists for powers of four.

Exercise 3 (Vector Field Visualization in ParaView, 9 Points)

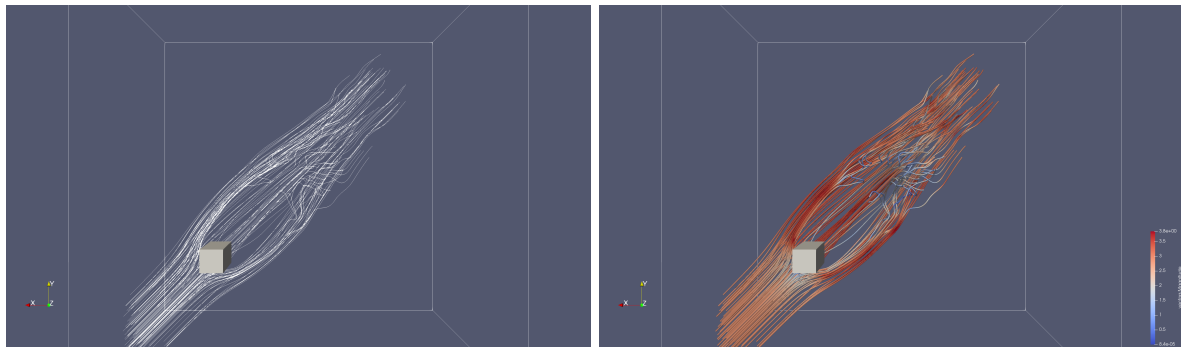


Figure 1: Left: Basic Streamlines. Right: Streamtubes with color-coded magnitude.

In this assignment, you will learn how to generate a geometry-based vector field visualization in ParaView.

- Please open the files `cuboid_velocity.vtk`, which contains a simulated flow around a cube, and `cube.vtp`, which represents the cube itself. Add a StreamTracer and set its seed type to “point source”. Compare results when seeding in front of the obstacle or behind it. Submit corresponding screen shots, one of which might look similar to Figure 1 (left). Do you see a reason to prefer one of the two seeding strategies over the other? (3P)
- Turn the unshaded line rendering into shaded streamtubes that provide a better impression of their 3D trajectories, e.g., via a Tube filter. Color the streamlines based on velocity magnitude, similar to Figure 1 (right). Submit a screenshot. Can you identify regions in which the flow speeds up or slows down? (3P)
- There are different reasons due to which VTK might terminate streamline integration. Find out which of them applies to most of your streamlines by color coding them according to **Reason-ForTermination** and looking up the interpretation of the respective numerical codes in the VTK documentation. How does your answer change when you substantially increase the maximum streamline length parameter? (3P)

Exercise 4 (Vector Field Visualization in VTK, 11 Points)

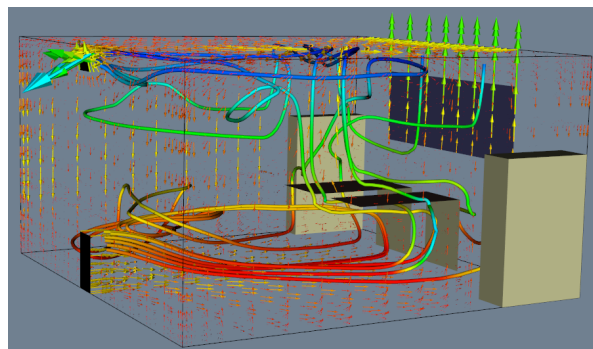


Figure 2: Stream tubes and arrow glyphs as you will render them in Exercise 2.

In this assignment, you will learn how to use VTK to perform basic geometry-based visualization of a flow field. You will need to download the files `stream-tubes.py` and `stream-data.vts` from eCampus.

The dataset contains a vector field that represents the air stream through an office room. The framework contains a partly implemented visualization pipeline, most of which is used to render furniture.

- By running the pipeline, you should see the office, with a cylindrical tube inside. Add a glyph visualization of the vector field using `vtkGlyph3D`. Connect it with `vtkXMLStructuredGridReader` as input and `vtkArrowSource` as source. Its output can be rendered as usual with `vtkDataSetMapper` and `vtkActor`. By default `vtkGlyph3D` will use the scalar data found on the grid to scale the arrows. Change the configuration so that it uses the magnitude of the vectors for scaling. (4P)
- In the pipeline, `SetStartPosition` is used to define the initial seed point. Replace it with a `vtkPointSource` to define six seed points in a region close to the door, similar to Figure 2. (3P)
- In the `StreamTracer`, set `InitialIntegrationStep` to $\{0.1, 0.4, 0.6, 1\}$, and run the experiment again for each of these values. How does this affect your result? In which regions are the changes most noticeable? Why? (2P)
- Repeat the same experiment after replacing the fourth-order Runge-Kutta scheme with the embedded scheme Runge-Kutta45. Is the effect from `InitialIntegrationStep` greater or smaller than before? Why? (2P)

Please provide your modified pipeline, the screenshot, and answers to the questions.

Exercise 5 (Numerical Integration, 13 Points)

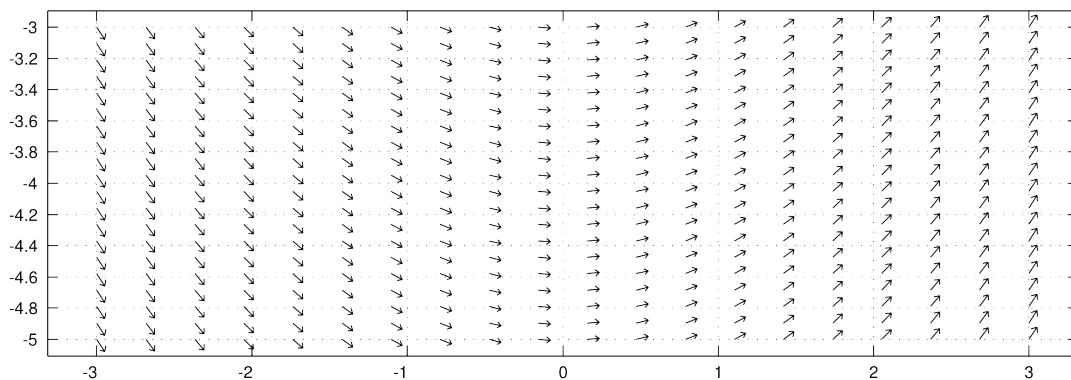


Figure 3: A vector field for numerical integration.

- Sketch three steps that the Euler method with a constant stepsize of $s = 1$ would take to integrate a streamline from the starting point $(x, y) = (-2.3, -3.4)$ in Figure 3. (2P) Note that, according to the axis labels, a step of size one will have approximately this length: $\longrightarrow$
- Sketch three steps that the second-order Runge-Kutta method with the same stepsize of $s = 1$ would take from the same starting point. (3P)
- Assume you repeatedly integrate a streamline of length l using Euler's method in the same vector field and from the same seed point, with decreasing step size s . With which asymptotic rate should the endpoints of these streamline converge towards the true endpoint of the continuous streamline? Why? (2P)
- Construct steady two-dimensional vector fields $\mathbf{v}(x, y)$ that have the following properties: With the same seed point and step size, Euler integration and Runge-Kutta integration...
 - produce exactly the same points. (2P)
 - produce different points, but they form the same streamlines. (2P)
 - produce different streamlines. (2P)

For each case, write down the equation for one example of a corresponding vector field.

Exercise 6 (Conservative Vector Fields, 7 Points)

By definition, a vector field $\mathbf{v}$ is conservative if it can be written as the gradient of some continuously differentiable scalar field ϕ , $\mathbf{v} = \nabla\phi$.

- a) Argue why any conservative vector field $\mathbf{v}$ is divergence-free, $\nabla \cdot \mathbf{v} = 0$, or provide a counter example. (2P)
- b) Argue why any three-dimensional conservative vector field $\mathbf{v}$ is irrotational, $\nabla \times \mathbf{v} = \mathbf{0}$, or provide a counter example. (2P)
- c) Which of the following two vector fields is conservative? How could you tell? Write down a corresponding scalar potential ϕ . (3P)

$$\mathbf{v}_1(x, y, z) = \begin{pmatrix} y^2 + z \\ x^2 - z \\ x^2 - y \end{pmatrix} \quad \mathbf{v}_2(x, y, z) = \begin{pmatrix} y + z \\ x - z \\ x - y \end{pmatrix} \quad (3)$$

Good Luck!